Applied Finance Project—5/3/2026

Niki Brooks, Kelly Hughes, JP Romero

Prof. Li

Machine Learning and LLMs (AIGB-724C-001)

I. Abstract

This project addresses a core inefficiency in equity research: a buy-side analyst covering a mid-cap stock manually has to pull three financial statements from a financial repository (SEC EDGAR, CapIQ, Bloomberg etc.), read 100+ pages of a 10-K, cross-check live price and sentiment data, overlay macro context from FRED, and synthesize it all into a memo. This process can take anywhere from 4 - 8 hours per company, and it's likely that the researcher might still miss important information, or fail to make useful connections with the data available.

The AI-Agent Stock Analyzer automates this workflow in under two minutes per ticker. The system uses a sector-routing classifier to direct each stock to one of eight specialist agents (REIT, Bank, Tech, Infrastructure, Energy, Consumer, Healthcare, Generic), each equipped with tailored system prompts and tools. Claude drives an autonomous tool-use loop against a retrieval layer grounded in SEC EDGAR XBRL facts and 10-K text, with every output enforced through a strict inline citation validator. The Analyzer focuses on REIT and Tech sectors, achieving the strongest RAG retrieval quality (80–100% section coverage), and includes brief coverage of other sectors like healthcare and consumer.

II. Data

Universe:

Scans 50 tickers at a time across 8 sector categories, with 10 REITs (O, PLD, AMT, SPG, EQIX, PSA, WELL, DLR, VICI, ARE), 10 infrastructure/EPC names (PWR, EME, FLR, KBR, DY, APG, MTZ, PRIM, AGX, ORA), and 5 names each in Banking, Tech, Energy, Consumer, Healthcare, and a Generic bucket covering diversified companies like BRK-B, GE, and BA. Any ticker outside the universe is classified on-the-fly via yfinance sector data and a Claude-based fallback.

Data sources and Time Periods:

Source	Type	Key Variables
SEC EDGAR XBRL	Structured financials	Revenue, EPS, leverage ratios, capital metrics — ticker-specific
SEC EDGAR 10-K	Unstructured text	Items 1, 1A, 2, 3, 7, 7A, 8 (Business, Risk Factors, MD&A, Financials)
yfinance	Live market data	Price, market cap, key stats, news headlines
FRED (public CSV)	Macro backdrop	Fed Funds rate, 10Y Treasury, CPI, unemployment
VADER	Sentiment	Headline compound score on Yahoo Finance news

EDGAR Ingestion

XBRL company facts are fetched via https://data.sec.gov/api/xbrl/companyfacts/CIK{cik}.json; full 10-K text via the EDGAR

submissions endpoint. Requests are rate-limited to 6/second (below the SEC's 10/second cap) and cached locally at data/cache/sec/. No API key is required.

Vector Store

ChromaDB with a PersistentClient, one collection per ticker ({ticker}_filings). 10-K text is chunked at 800 characters with 150-character overlap, respecting section ("Item X") boundaries. Embeddings use sentence-transformers/all-MiniLM-L6-v2. This section-aware chunking is a deliberate design choice to keep MD&A text from bleeding into Risk Factor retrieval.

Cleaning and Processing

XBRL tags are parsed into structured fact dictionaries and fed directly to Claude as tool call results. 10-K text is extracted using regex-based Item markers, which works well for standard filings but fails on non-standard formats (known failure: BRK-B). VADER sentiment is computed on Yahoo Finance headlines and aggregated as a compound score at the ticker level.

III. Applied Finance (Core Section)

Financial Problem

Can agent-based RAG systems leverage SEC filings and live market data to produce research memos that are analytically correct, sector-specific, and fully traceable to their respective sources, automating the research process for junior analysts?

Framework (Sector-Specialized Agents)

Each sector requires analysts to use structurally different approaches to investigate their corresponding data e.g. REIT memos need FFO, AFFO, cap rates, and occupancy while bank memos need information on NIM, Tier 1 capital, and credit quality. Rather than using a single generic agent, the system routes each ticker to a specialist with a tailored system prompt and custom tools. The classifier uses yfinance sector/industry metadata combined with keyword rules and a Claude-based LLM fallback for ambiguous cases.

Framework (Claude Tool-Use Loop)

The orchestrator (agents/orchestrator.py) runs a true agentic loop: Claude decides which tool to call next, rather than following a scripted pipeline. The tool set includes 12+ shared tools covering EDGAR XBRL retrieval, 10-K section search, live price/sentiment fetch, FRED macro pull, and bull_prob scoring, plus specialist-specific tools per sector. This autonomy means Claude can follow unexpected paths e.g. if an initial XBRL query returns missing data, it can fall back to the 10-K text retrieval on its own.

Framework (RAG and citation enforcement)

The retrieval layer surfaces relevant 10-K passages using ChromaDB semantic search. The citation validator (evaluation/citation_validator.py) then post-processes every generated memo: it extracts every dollar amount, percentage, and ratio in the text and checks that each has a citation marker within 150 characters. Memos with less than 70% citation coverage are rejected. This is the backbone of the system's core value proposition — traceability. Every numerical claim can be audited back to a specific SEC filing, XBRL tag, or live data source.

Bull Probability Scoring

Rather than the pre-trained static Random Forest used in the initial attempts, this system computes a live composite bull_prob per ticker from momentum, revenue/earnings growth signals, VADER sentiment, and leverage metrics. This is heuristic rather than backtested, but transparent and responsive to current data.

Evaluation Results

Retrieval quality was assessed against a 34-question golden set (evaluation/golden_questions.json) covering analyst-style queries across all 8 sectors. Each question specifies expected tools, expected 10-K sections, required regex patterns, and minimum citation count.

Overall, 18/33 cases hit at least one expected section (54% weighted hit rate), with a mean section coverage of 34.8% and mean relevance score of 0.331 across the full universe. The end-to-end answer quality evaluation uses a three-part pass gate: completeness ≥ 67%, citation coverage ≥ 70%, and tool coverage ≥ 50%. The baseline REIT test (Realty Income, question reit-001) passed all three: 100% completeness, 75% citation coverage, 50% tool coverage.

Additionally, the pattern in retrieval performance maps cleanly to 10-K structure. REITs and large-cap tech companies write dense, well-organized 10-Ks with consistent section headers and plain prose, ideal for sentence-transformer embeddings. Healthcare 10-Ks are dominated by regulatory/clinical language (pipeline, LOE, PDUFA) that diverges semantically from the query language a financial analyst would use. This is not a model failure; it is a domain adaptation gap, fixable with sector-specific query rewriting. The BRK-B failure is a known edge case: Berkshire uses a non-standard 10-K format that breaks regex-based Item extraction.

Regarding LLM usage, Claude (claude-sonnet via the Anthropic API) drives the full agentic loop; tool selection, synthesis, and memo generation. The specialist system prompts define sector-appropriate memo sections and explicitly inject the citation rules from agents/specialists/_shared.py. The citation validator provides a hard quality gate that the LLM

cannot circumvent through confident-sounding but unsourced claims, which is the key accountability mechanism the system adds over a naive chatbot.

IV. Conclusion

The AI-Agent Stock Analyzer demonstrates that a sector-specialized, citation-enforced agentic architecture can compress a 4–8 hour analyst research workflow to under two minutes per ticker, with every numerical claim traceable to an SEC source. The REIT and Tech specialists are production-ready on the current evaluation set. The most significant finding is the structural link between 10-K quality and RAG performance: the system works best where filings are consistently structured and worst where domain-specific language creates a semantic gap between filing content and analyst query language. This is a clear, actionable diagnosis.

The most important limitations are the Healthcare retrieval gap (0/4 hit rate, needs sector-specific query rewriting), the absence of 10-Q support (missing quarterly changes), and the fact that bull_prob is a heuristic composite rather than a backtested return predictor. The citation threshold of 70% also leaves room for uncited claims that a real compliance process would flag.

An analyst using this system today for REIT or large-cap tech coverage could plausibly cut initial screening time by 80%+, using the generated memo as a draft that they verify and annotate rather than build from scratch. For healthcare or non-standard filers, the system in its current form is better used as a data-pull assistant than a memo generator.